

A systematic comparison of methodologies for the estimation of the serial interval

Lanlan Yu ^{a,b}, Yunyi Cai ^{a,b}, Gui-Quan Sun ^{c,d}, Jiancheng Lv ^{a,b}, Quan-Hui Liu ^{a,b,*}

^a College of Computer Science, Sichuan University, Chengdu, China

^b Engineering Research Center of Machine Learning and Industry Intelligence, Ministry of Education, Sichuan University, Chengdu, China

^c Department of Mathematics, North University of China, Taiyuan, China

^d Complex Systems Research Center, Shanxi University, Taiyuan, China

ARTICLE INFO

Article history:

Received 8 April 2025

Received in revised form 3 March 2026

Accepted 9 March 2026

Available online 10 March 2026

Handling editor: Daozhou Gao

Keywords:

Serial interval

Epidemic modeling

Serial interval estimating methods

ABSTRACT

Serial interval (SI) is a crucial indicator for characterizing the course of epidemic transmission and provides guidance in designing the intervention strategy for preventing epidemic spreading. Many approaches have been developed to estimate the serial intervals, while their effectiveness evaluations are rarely exhibited due to the unavailable of the real SI values. Here, we developed the data-driven based transmission model to simulate the “realistic” epidemics and kept the entire micro-transmission chain to obtain the real SI values for comparison. We firstly found the direct statistic of serial interval obtained from the infector-infectee pairs within households can serve as an accurate representative of the entire population. The households monitored during the early and peak phases of an epidemic provide precise estimations, while the serial interval would be underestimated if the households are collected during the decreasing phase of epidemic. This study reveals that both the *ML* method using the contact tracing data and the *BY* method imputing with the household follow up data outperform that the *TM* method which employing the aggregated data during the early and decreasing phases of epidemic, while the *TM* method provides a more precise estimation of mean value of SI during the peak phase of epidemic. Our study also presents that sampling 100 households would be enough to give an accurate estimation of SI value. Other sensitiveness on both the epidemic transmission parameters and the noise in the date of symptom onset demonstrate the robustness of the above results. This study provides a systematic comparison of SI estimation methods and gives the public health authority the evidence when they are using the relevant SI estimating methods to inform the policy.

© 2026 Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

The serial interval (SI), defined as the difference between the time of symptom onset of secondary case and of their primary infector (Ali et al., 2020; Nishiura et al., 2020; Tuite et al., 2010; Vink et al., 2014), is a crucial indicator for

* Corresponding author. College of Computer Science, Sichuan University, Chengdu, China.

E-mail address: quanhui.liu@scu.edu.cn (Q.-H. Liu).

Peer review under the responsibility of KeAi Communications Co., Ltd.

understanding the transmission dynamics of infectious diseases (Dhiman et al., 2024; S. W. Park et al., 2024). SI is instrumental to guide public health response efforts, including the design of contact tracing guidelines, a time window for monitoring symptoms in contacts of identified cases (Paquet et al., 2006), and an indicator for evaluating the effectiveness of control interventions (Ali et al., 2020; WHO Ebola Response Team et al., 2014). SI also plays a key role in understanding transmission patterns (Nash et al., 2023). However, estimating SI can be challenging as the infector-infectee pair (i.e., who infects whom) can rarely be observed directly and is usually based on the mere knowledge of the presence of an epidemiological link. Moreover, SI vary across contexts due to differences in implemented public health interventions, human behavior, and social mixing patterns, thus differences in SI can be observed also during different phases on an epidemic (Madewell et al., 2023; M. Park et al., 2020; Rai et al., 2021; Vink et al., 2014). Given the importance of SI and the challenges in estimating its distribution, a set of approaches have been introduced to estimate SI from empirical epidemiological data (Chen et al., 2022; Forsberg White & Pagano, 2008; S. W. Park et al., 2021; te Beest et al., 2013; Vink et al., 2014).

Different approaches for the estimation of SI have been proposed, each one relying on a set of assumption and type of available data. When contact tracing data is available, statistical methods have been developed to identify possible infector-infectee pairs and estimating the distribution of SI by exploring the likelihood of the reconstructed transmission pairs (Chen et al., 2022; Du et al., 2020; Li et al., 2021; S. W. Park et al., 2021; te Beest et al., 2013; Yang et al., 2020). Lessler et al. (Lessler et al., 2009) developed a systematic approach to identify transmission pairs and estimated the SI distribution by fitting parametric models, such as the Weibull and log-normal distributions. Similarly, te Beest et al. (te Beest et al., 2013) applied a maximum posterior probability approach to reconstruct transmission pairs and estimated the SI distribution for influenza H1N1 across different social settings, including households, schools, and workplaces.

For household data, a common approach is to define a case-to-case interval (index-case-cases, ICC) based on the assumption that the first reported case in a household is the index case (Cauchemez et al., 2009; Cowling et al., 2009; Leung et al., 2011; Levy et al., 2013; Orellano et al., 2012; Savage et al., 2011). Distinct from the SI, ICC intervals (index-case-cases) refer to the empirically observed time gap between the first case (index case) and the sequentially reported secondary cases in the same household (Vink et al., 2014), regardless of the specific infector-infectee link. By characterizing the link between ICC intervals and SI of potential infector-infectee pairs, Cauchemez et al. analyzed data from 216 households with an index patient during the 2009 H1N1 influenza pandemic in the US, estimating a mean SI of 2.6 days (2009). Other studies estimated the mean household SI directly from or by fitting distribution to ICC intervals (Cauchemez et al., 2009; Levy et al., 2013; Susvitasari et al., 2023; Vink et al., 2014). Vink et al. (Vink et al., 2014) accounted for various transmission pathways within households, such as two cases sharing the same coprimary source and allowing for negative serial intervals (i.e., the infector has symptom onset after that of the infectee), using a probability model to estimate the SI distribution for influenza. Susvitasari et al. (Susvitasari et al., 2023) extended the ICC interval concept to adjust for partially sampled data in larger social settings beyond households.

When only aggregated data is available, White et al. developed a maximum likelihood method to simultaneously estimate the reproduction number and SI distribution, which they applied to influenza data (Forsberg White & Pagano, 2008; White et al., 2009). Similarly, Archer et al. imputed missing data from the first wave of the influenza A (H1N1)pdm09 pandemic in South Africa and estimated its SI (Archer et al., 2012). Building on these statistical frameworks, Dai et al. (Dai et al., 2023) developed a likelihood-based approach to jointly estimate the instantaneous reproduction number $R(t)$ and SI distribution.

The aim of this study is to carry out a systematic comparison of several widely adopted methodologies for the estimation of SI (Dai et al., 2023; te Beest et al., 2013; Vink et al., 2014) in a controlled setting for which the ground truth is known. Specifically, we developed an individual based model of influenza transmission calibrated that has been validated against empirical epidemiological and seroepidemiological data that keeps track of the entire transmission chain. The model was used to create synthetic patient line lists, including information on patients' household contacts, which served as input for the four approaches to estimate SI. Performance of the methodologies were assessed against the ground truth coming from the simulation for different levels of data completeness and accuracy, epidemic phases, pathogen transmissibility, risk of developing symptoms upon infection, and transmissibility of asymptomatic individuals relative to symptomatic ones.

2. Methods

2.1. Pathogen transmission model and synthetic dataset

2.1.1. Multilayer network

We used a synthetic population of about 500,000 individuals representing a subsample of the Italian population taken from our previous studies (Liu et al., 2018, 2022). Everyone in the synthetic population is assigned to a household and to a general community. Moreover, according to employment rates by age, individuals are assigned to workplaces, while students are assigned to schools according to school enrollment rates by age. This synthetic population allows us to define a multilayer network of contacts (see Fig. 1A) consisting of the household (H), school (S), workplace (W), and community (C) layers, which represent the main settings for influenza transmission (Ferguson et al., 2005; Merler & Ajelli, 2010). Further details regarding multiplex network are provided in the Supplementary Material.

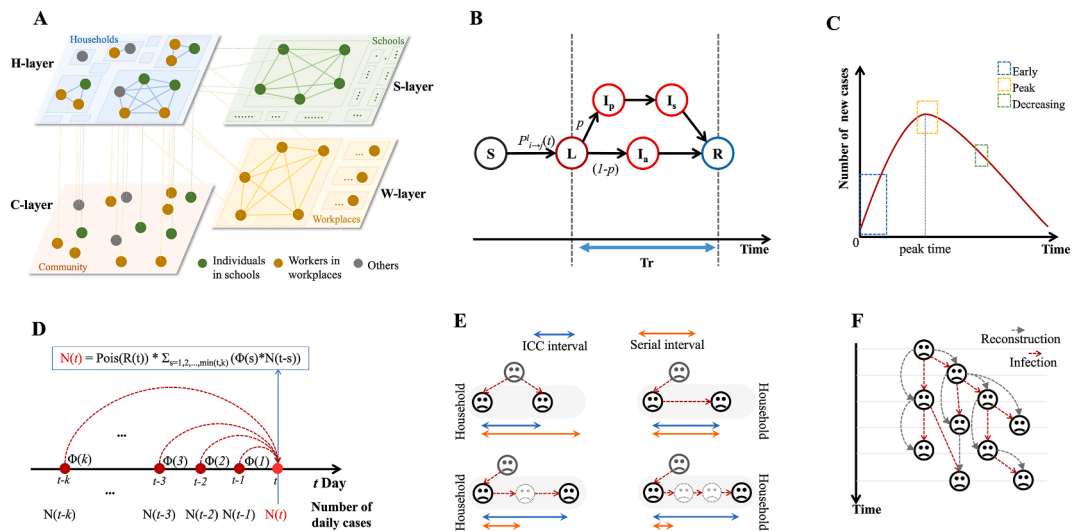


Fig. 1. The schematic of influenza transmission model and the serial interval estimating methods.

A The schematic of multi-layer contact network. **B** The flow of states in transmission model. **C** The visualization of three monitoring scenarios. **D** The diagram of assumption in estimating SI methods using aggregated data. **E** The schematic of four relationships within household between SI and ICC intervals, as proposed by Vink et al. (Vink et al., 2014). **F** The diagram illustrating contact tracing pairs with infection and reconstruction.

2.1.2. Influenza transmission model

We developed a stochastic agent-based model of influenza transmission on the multiplex network. Influenza transmission is simulated according to an SLIR (susceptible, latent, infectious, removed) scheme (Lee et al., 2016; Liu et al., 2022), where infectious individuals were further divided into pre-symptomatic (I_p), symptomatic (I_s), and asymptomatic (I_a) individuals (Fig. 1B). If a susceptible individual i is connected on the multilayer network with an infectious (either pre-symptomatic, symptomatic, or asymptomatic) individual j , susceptible individual i will be infected by infectious individual j with a layer-specific probability. The infectiousness of an infected individual follows a temporal trajectory following a gamma function $\kappa(\tau)$ (Ferguson et al., 2005). When symptomatic and asymptomatic individuals are no longer infectious, they acquire immunity that lasts for the entire duration of the simulated outbreak.

The influenza transmission probability between a pair of infectious and susceptible individuals accounts for four factors: age-specific susceptible to infection (Fraser et al., 2009), layer-specific per-contact transmission risk (Liu et al., 2018), the epidemiological status of the infectious individual (I_p , I_s , or I_a), and the time elapsed since infection (i.e., temporal infectiousness profile) (Ferguson et al., 2005). The incubation period is assumed to follow a gamma distribution with mean equal to 1.6 days (shape = 2.62, scale = 0.42 and shifted by 0.5) truncated at 10 days (Ferguson et al., 2005). The probability of developing symptom is assumed to be 0.5 (Ferguson et al., 2005; Merler & Ajelli, 2010). We calibrated the weights of the contacts in each layer such that the fraction of infections associated with household, school, workplace, and community contacts is 30%, 18%, 19% and 33%, respectively, as reported in influenza literature (Ferguson et al., 2005; Liu et al., 2018; Merler & Ajelli, 2010). Further, we estimated the age-specific susceptibility to infection by calibrating the model to match the seropositive rates by age at the end of the 2009 H1N1 pandemic in Italy (Merler et al., 2013). Details on model parameters and the calibration procedure are reported in the Supplementary Material.

2.1.3. Generated synthetic line lists

Our model produces a patient line list comprising information on the date of symptom onset, primary infector, list of household contacts for each synthetic agent. Then, we sample n households ($n = 100$ in the baseline analysis) with at least 2 members. Using these households, we compile a dataset containing all the members of these households with the ID of the synthetic individual, the ID of the household, and the date of symptom onset or NA if that individual was not infected or did not show symptoms.

2.2. Methods for the estimation of the serial interval

We implemented three widely adopted approaches for the estimation of serial interval (Dai et al., 2023; te Beest et al., 2013; Vink et al., 2014), and a no-priori-assumption statistical approach. Specifically.

- 1) Dai et al. (Dai et al., 2023) (here referred to as *TM*) utilized the assumption that the daily number of new cases $N(t)$ at time t can be approximated by a Poisson (Ajelli et al., 2014, 2016) (see Fig. 1D) and construct the following likelihood function:

$$L = \prod_{t=1}^T \frac{e^{-R(t)N(t)} (R(t)N(t))^{N(t)}}{N(t)!}, \quad (1)$$

where,

$$N(t) = \sum_{i=1}^{\min(k,t)} N(t-i)\Phi(i). \quad (2)$$

Therein, $R(t)$ is the instantaneous reproduction number at day $t = \{1, 2, 3, \dots, T\}$, $\Phi = \{\Phi(1), \Phi(2), \dots, \Phi(k)\}$ is the probability vector of SI distribution, $N(t)$ is the daily number of cases at day t . Then they employed the Bayesian information criterion (BIC) and the trust-region algorithm to maximum the likelihood function, as well as to estimate $R(t)$ and SI distribution.

2) Vink et al. (Vink et al., 2014) (here referred to as *BY*) accounted for four possible transmission pathways to develop the relationship between SI and ICC interval (see Fig. 1E) and to construct the likelihood function:

$$L(w_1, w_2, w_3, \mu, \sigma^2 | \mathbf{x}) = \prod_{i=1}^C P(t_i; w_1, w_2, w_3, \mu, \sigma^2), \quad (3)$$

$$P(t) = \int_{t-1}^{t+1} f(x)g(x)dx, \quad (4)$$

$$f(x|\mu, \sigma^2) = w_1 f_1(x|\sigma^2) + w_2 f_2(x|\mu, \sigma^2) + w_3 f_3(x|\mu, \sigma^2) + (1 - w_1 - w_2 - w_3) f_4(x|\mu, \sigma^2) \quad (5)$$

Here, $\mathbf{x} = \{t_1, t_2, t_3, \dots, t_C\}$ is the vector of ICC intervals. μ and σ^2 are mean and variance of SI distribution (i.e. normal distribution), respectively. $g(x)$ is the triangular distribution function, and $f(x|\mu, \sigma^2)$ is the mixture probability distribution for ICC interval. $f_i(x)$ is the component density for the i 'th transmission path, where $i = \{1, 2, 3, 4\}$. Furthermore, they employed the robust expectation-maximization algorithm to estimate the parameters ($w_1, w_2, w_3, \mu, \sigma^2$).

3) Te Beest et al. (te Beest et al., 2013) (here referred to as *ML*) distinguished the exact onset time and the reported day, and then applied the Markov chain Monte Carlo method (MCMC) to maximize the likelihood function which is following:

$$L(\theta | \mathbf{x}) = \prod_{i=1}^H f_i(\theta | \mathbf{x}(i)), \quad (6)$$

$$f_i(\theta) = \prod_{j=1}^C P(t_j | \theta) \quad (7)$$

Therein, $\mathbf{x}(i) = \{t_1, t_2, t_3, \dots, t_C\}$ is the SI vector for a possible transmission pathway within i 'th social setting, where $i = \{1, 2, 3, \dots, H\}$ and $C + 1$ is the number of symptomatic members here. $P(t_j | \theta)$ is the probability for SI = t_j corresponding to an infector-infectee pair. Here, only with the onset time, we adjusted equation (7) into $f_i(\theta) = \max P(t_j | \theta)$ to simplify the approach and to accelerate convergence of MCMC. Specially, $P(t_j | \theta)$ is initially assigned with a norm distribution with parameters $\theta = \{\mu, \sigma^2\}$.

4) No-priori-assumption statistical approach (here referred to as *MI*) is designed as a multiple imputation reference approach without assumption about transmission pathway. We randomly selected a possible transmission pathway within household and calculated serial interval for this configuration. Here we repeated the process described above 50 times to estimate SI distribution.

Details are reported in the Supplementary Material.

2.3. Evaluation metrics and analyzed scenarios

To systematically compare the methodologies for the estimations of the serial interval, we compared the estimated serial intervals with the ground truth using the following four metrics:

1) The difference of mean SI (D_{mean}) between the SI distribution estimated by the method and the ground truth distribution.

- 2) The maximum distance (D_{KS}) between the cumulative SI distributions estimated by the method and the ground truth distribution, as measured using the Kolmogorov-Smirnov test.
- 3) The Total Variation Distance (D_{TVD}) between the SI distribution estimated by the method and the ground truth distribution.
- 4) The Jensen-Shannon Divergence (D_{JSD}) between the SI distribution estimated by the method and the ground truth distribution.

According to the definition of the four metrics above, smaller values indicate the better performance of the estimating methods. Details on these metrics are provided in the Supplementary Material.

To explore the performance of the four analyzed SI estimation methods, we considered three phases of an epidemic outbreak: the early phase, peak phase, and decreasing phase (see Fig. 1C). Details are exhibited in the Supplementary Material. For the early phase we sample the first n infected households with at least 2 members and one of them is the symptomatic primary infector; for the peak phase, we sample n infected households during the epidemic peak; and for the decreasing phase, we sample n infected households from the population 2 weeks late after the epidemic peak. In the baseline analysis, we set the basic reproduction number $R_0 = 1.5$, the probability of individual developing symptom $p = 0.5$, the relative infectiousness of the asymptomatic individual $q = 1.0$, the number of sampled households $n = 100$, and the no-noise in onset time of symptomatic case. In the meanwhile, we explored $R_0 = \{1.3, 1.5, 1.7\}$, $p = \{0.3, 0.5, 0.7\}$, $q = \{0.3, 0.57, 0.7, 1.0\}$, $n = \{50, 100, 200, 400\}$ and the bias onset time of cases (gauss noise with $\mu = 0, \sigma^2 = 1$) as the sensitivity analyses.

3. Results

In the experiment, we simulated influenza transmission starting with one seed for 1000 simulations. All the following results are averaged or evaluated by using these simulations of epidemic outbreak.

3.1. Statistical analysis

The daily effective reproduction number $R(t)$ is defined as the average number of secondary cases generated by an infectious individual at time t (Liu et al., 2018), which is an indicator as important as the serial interval (SI) in understanding and characterizing the evolution of the emerging infectious disease. We firstly reported $R(t)$ and the time evolution of serial interval in Fig. 2A and B, which are directly computed from the micro-simulations by keeping track of the number of secondary infections generated by each individual infected at time t . We found that $R(t)$ increases with time in the early phase of the epidemic, starting from $R_0^{index} = 1.5$ to the peak of about 1.7, which is similar to the pattern as observed in the data-driven model of epidemic in (Liu et al., 2018). While the average of serial interval decreases over the whole time, this differs from the pattern observed in reference (Liu et al., 2018) which reaches the shortest right before the peak time. It might be due to the time varying of infectiousness of infected individuals in the current model.

We calculated the serial interval (SI) using the infector-infectee pairs from two data sources: all transmission pairs occurring in the households (Household-level) and those from the entire population (Population-level) which include all transmissions occurring in households, schools, workplaces and community. As shown in Fig. 2C, the mean values of SI at the Household-level and the Population-level are respectively as 2.68 days (median: 2.75, Q1: 1.75, Q3: 3.5) and 2.67 days (median: 2.75, Q1: 1.75, Q3: 3.5). They are very closer, and in the following, we employed the distribution of SI obtained from the Population-level as the ground truth for assessing the effectiveness of the SI estimating methods.

Additionally, we directly computed the SI using the symptom onset times of the infector and infectee of the transmission pairs observed in the households monitored in different epidemic phases, as shown in Fig. 2D. We found the mean values of SI calculated from the households monitored during the early and peak phases are very close to the ground truth, but a wider confidence interval is observed in the peak phase. While the obtained mean value of SI in the decreasing phase is smaller than the ground truth.

3.2. Evaluation of SI estimating methods

We found that the *TM*, *BY*, *ML*, and *MI* methods overestimated the serial interval (SI) during the early phase of the epidemic, as shown in Fig. 3A. Among these methods, the *MI*, *ML*, and *BY* approaches estimated the SI mean values closer to the ground truth compared to the *TM* method, which is further supported by their evaluation across other metrics (i.e., D_{TVD} , D_{JSD} , and D_{KS}), as illustrated in Fig. 3D. Similar to the early phase, all the four methods overestimated the SI mean during the peak phase of the epidemic. While the *TM* method achieved the most accurate SI mean value in this phase (Fig. 3B), the *MI*, *ML*, and *BY* methods outperformed the *TM* method in terms of the other three evaluation metrics, as shown in Fig. 3E. In the decreasing phase of the epidemic, the *MI*, *ML*, and *BY* methods provided the most accurate SI estimates, as presented in Fig. 3C and F. In a summary, the SI estimating method using the contact tracing data (i.e., *ML*) and the household follow up data (i.e., *BY*) perform better than the method *TM* which only uses the aggregated data during the early and late phases of epidemic, while the latter provides the precise estimation of mean value of SI during the peak phase of epidemic.

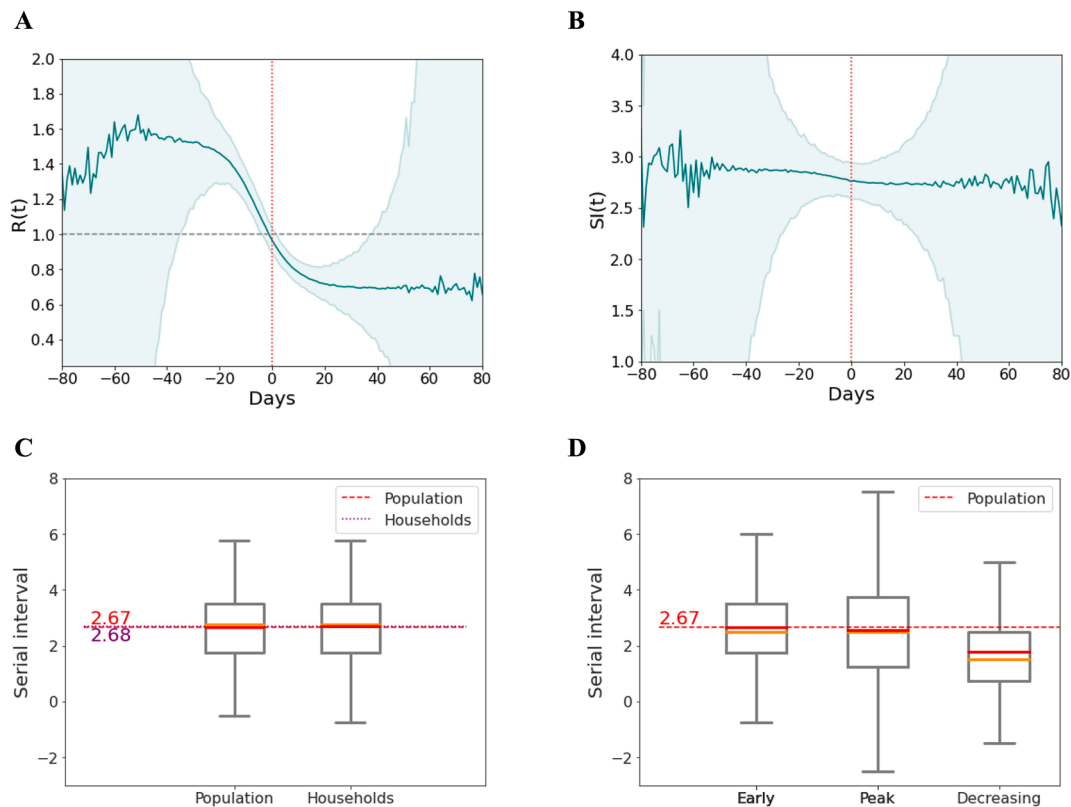


Fig. 2. The result of statistical analysis.

Panels **A** and **B** respectively show the effective reproduction number ($R(t)$) and the serial interval (SI) versus time. In detail, the dark green line represents the average, while the light green lines correspond to the 97.5% and 2.5% percentiles, respectively. Panel **C** presents the obtained serial interval (SI) using infector-infectee pairs occurring in the entire population (Population-level) and only occurring within households (Household-level). Panel **D** shows the SI using infector-infectee pairs sampled from households in different epidemic phases. In the boxplot, the red short line indicates the mean value, and the orange short line represents the median value. While the red and purple dotted lines correspond to the mean serial interval at the population level and household level, respectively.

3.3. Sensitiveness on the number of monitored households

Determining the minimum number of monitored households to achieve an accurate estimation of the serial interval (SI) is crucial for epidemic surveillance. As shown in Fig. 4, the impact of number of households on the SI estimating methods is evaluated in terms of four metrics under three monitoring scenarios. We observed that with the increase of the number of monitored households during the early phase of epidemic, the performance of all these four SI estimating methods is improved, as shown in Fig. 4A, D, 4G, and 4J. To be noted that, the performances of the MI , ML , and BY methods almost stabilized when the number of households reached or exceeded 100. In contrast, the TM method showed gradual improvement as the sample size increased, except for the peak phase (presented in Fig. S5 of the Supplementary Material). These trends persisted during the epidemic's peak and decreasing phases, as illustrated in Fig. 4.

Across all epidemic phases, the MI , ML , and BY methods demonstrated comparable performance in terms of the D_{mean} and D_{KS} metrics. Moreover, the ML and BY methods produced identical results when evaluated using the D_{TVD} and D_{JSD} metrics. While the MI method was more sensitive to the number of available households, particular in terms of the D_{KS} , D_{TVD} and D_{JSD} metrics, it achieved comparable effectiveness to the ML and BY methods when the sample size exceeded 100 households.

3.4. Sensitiveness on the parameters of epidemic transmission

In this section, we provided the systematic comparisons and evaluations for the robustness of the SI estimating methods under a variety of sensitivities experiment, by varying the parameters related to influenza transmission and symptom onset timing.

As illustrated in Fig. 5, we found increasing the basic reproduction number (R_0) improves the performance of TM method in the early phase but weakens its performance in the peak phase through all the four metrics. In the meanwhile, the performance of MI method is lifted during the early and peak phases of epidemic in terms of the metrics D_{KS} , D_{TVD} and D_{JSD} .

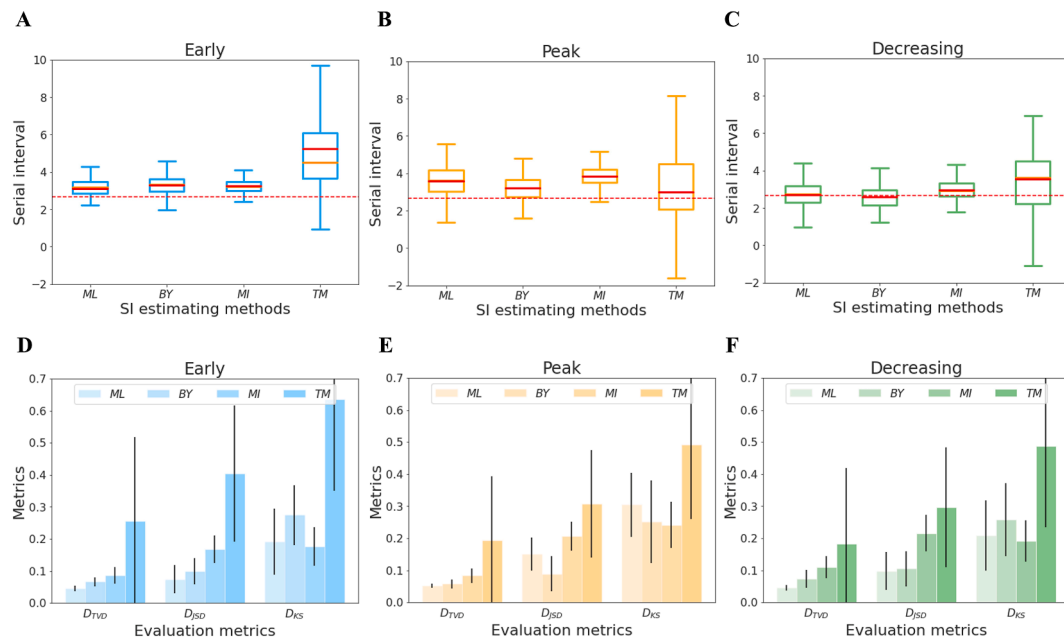


Fig. 3. The evaluation of SI estimating methods.

Panels A–C present the SI estimations of four approaches during the early, peak and decreasing phases of epidemic, respectively. The red dashed line in panels A–C denoted the mean of ground truth SI, the red short line in boxplot represents the mean SI, while orange short line represents the median SI. Panels D–F present the quantitative difference in terms of D_{TVD} , D_{JSD} , and D_{KS} , between the estimated SI and ground truth using four estimating methods under the early, peak and decreasing phases of epidemic, respectively. Here, black vertical line is error bar. 100 households are employed in the estimation of SI for all these four approaches.

Increasing the probability of developing symptoms (p) improves the estimation accuracy consistently for all four SI estimating methods under three monitoring scenarios in terms of four metrics. Conversely, increasing the relative infectiousness of the asymptomatic individual (q) diminishes the effectiveness of SI estimating methods. These results could be explained that the former would increase the number of available infector-infectee pairs for the SI estimation, while the latter would decrease such number. We found incorporating the noise to the time of symptom onset could decrease the effectiveness of all the SI estimating methods. Lastly, as shown in Fig. 5, the ML and BY methods perform best across all monitoring scenarios in all metrics, while TM performs worst.

In a summary, although the different effects of the sensitiveness for the four SI estimating methods are observed. We found the MI and TM methods are a little more sensitivity to the change of the basic reproduction number compared with the BY and ML methods. Increasing the number of available symptomatic infector-infectee pairs will enhance the estimation accuracy.

4. Discussion

Serial interval is an important indicator for characterizing the speed of epidemic spread and providing the guidelines for intervention strategy designing. Although there are many studies have developed the approaches to estimate the serial interval, the effectiveness evaluation of these estimating methods is lack due to the absence of real SI information, the various factors in epidemic transmission, and the differences in household-based surveillance methodologies (Kenah et al., 2008; Liu et al., 2018; Madewell et al., 2023; M. Park et al., 2020; Rai et al., 2021).

In this paper, we developed the data-driven based transmission model to simulate the influenza epidemics and kept the entire micro-transmission chain to enable ground-truth SI calculation for comparison. Through comprehensive literature review, we implemented three established SI estimating methods along with an assumption-free approach to carry out a systematic methodological comparison. Meanwhile, we designed three kinds of monitoring scenarios, listed as the early, peak and decreasing phases of epidemic to collect data and evaluated the SI estimating methods in terms of the mean difference, the maximum distance, the total variation distance and the Jensen-Shannon divergence between the estimated SI distribution and the distribution of ground truth.

Our study presented that SI estimation approach performs better when more detailed epidemic data is available. When only aggregated data is available, the SI estimating method (i.e., TM) assuming the observed number of symptomatic cases at each day follows a Poisson, and thus establishes the relationship between this number with the serial interval distribution and the reproduction number characterized by equation (1) (He et al., 2020). This method would not be able to capture the

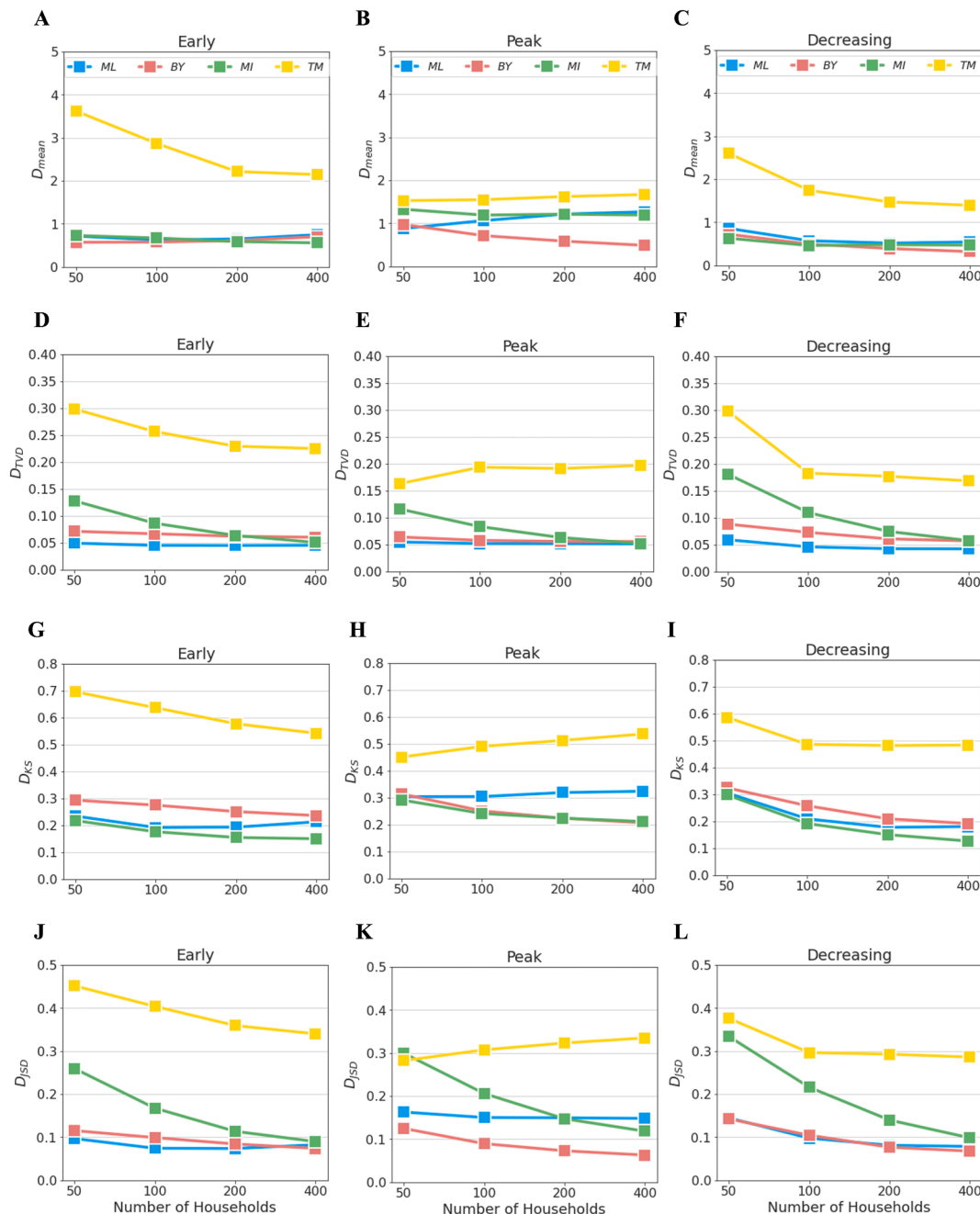


Fig. 4. The sensitiveness analysis of SI estimating methods on the number of sampled households.

Panels **A–C** respectively show the performance of SI estimating methods evaluated by using the difference of mean (D_{mean}) between the estimated SI distribution and the ground truth distribution during the early, peak and decreasing phases. Similar to panels **A–C**, panels **D–F** show the performance of SI estimating methods evaluated by using the Total Variation Distance (D_{TVD}), panels **G–I** present the performance of SI estimating methods evaluated by using the maximum distance (D_{KS}), panels **J–L** present the performance of SI estimating methods evaluated by using the Jensen-Shannon Divergence (D_{JSD}).

transmission caused by the pre-symptomatic individuals, and also exhibits the bias arising from the assumption of constant infectiousness of the infected individual (Archer et al., 2012; Dai et al., 2023; Forsberg White & Pagano, 2008; White et al., 2009). These thus lead to the TM method (Dai et al., 2023) underperforming MI method in the simulated epidemic data. In comparison, when individual onset times are available, both SI estimation approaches, whether based on assumptions or not, would perform better. These methods (i.e., MI, ML and BY) are with the similar performance in terms of the estimation for the mean SI. Therein, assumption-based approaches (i.e., ML and BY) outperform method without assumption (i.e., MI). In specific, the BY method (Vink et al., 2014), based on the household follow-up data, regards the first symptomatic case in

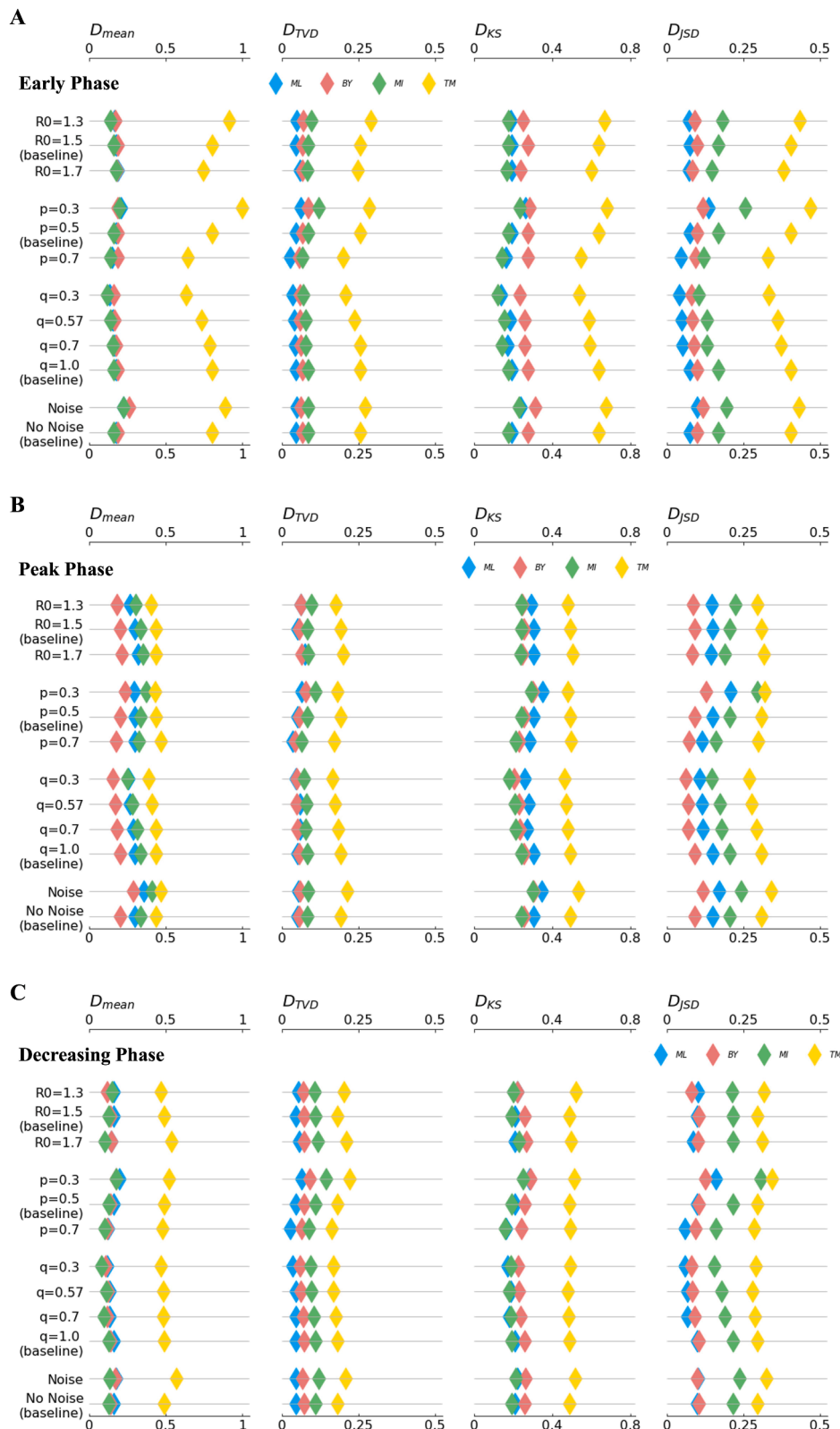


Fig. 5. The sensitiveness analysis of SI estimating methods on influenza transmission parameters and noise of symptom onset time evaluated by using four metrics. Panels **A–C** show the performance evaluated by D_{mean} , D_{KS} , D_{TVD} and D_{JSD} at three phases of an epidemic outbreak, respectively. The influenza transmission parameters include the basic reproduction number (R_0), the probability of symptom onset (p), the relative infectiousness of asymptomatic individual to symptomatic individual (q). We also assumed the date of symptom onset is with a noise characterized with the Gaussian distribution with a mean of zero and a standard deviation of 1.

the household as the only source of infections (Cauchemez et al., 2009; Cowling et al., 2009; Levy et al., 2013; Savage et al., 2011), and develops the probability model accounting for various transmission pathways within households to estimate the SI distribution. This method presents to be effective and robust during various monitoring scenarios. While the method (i.e., *ML*) utilizes the maximum likelihood to reconstruct the transmission pairs (Du et al., 2020; te Beest et al., 2013), rather than assigning transmission pairs (such as *BY* and *MI*). Therefore, *ML* commonly performs similarly to *BY* method (with adaptive mixture model about transmission pathways) but better than *MI* method (without assumption about reasonable transmission pairs). Furthermore, when contact tracing data is available and accurate, estimation approach, according to the entire transmission chain within household, can calculate the “realistic” SI value in the early phase of epidemic.

The sensitivity analysis of the SI estimating methods covers the parameters from the influenza transmission model, the noise in symptom onset timing and the size of monitored households. We found varying the parameters would increase the number of the possible infector-infectee pairs, which could improve the estimation of SI. This also happens to enlarge the size of monitored households. Our results also reveal that monitoring 100 households would be enough to achieve robust SI estimations for influenza transmission, though this threshold remains sensitive to the density of possible infector-infectee pairs in data-sparse environments. Notably, all these results are robust to the noise in the symptom onset timing.

Although this study provides a systematic comparison of methodologies for the estimation of the serial interval. There still exhibits the following limitations: 1) In this study, we evaluated the SI estimating methods based on the simulated epidemic data by simulating the “realistic” spreading of influenza. Our data-driven model was carefully calibrated with empirical surveillance data, while it remains an approximation of actual disease spread. It should be cautious when interpreting them to inform the policy. 2) The effectiveness evaluations of these SI estimating methods are limited to the influenza like disease. Future work should explore how these results generalize to other infectious diseases or varying strains of respiratory pathogens, which may exhibit distinct transmission profiles (Xiong et al., 2024). 3) Although, we classified the SI estimating methods into three categories based on the data used and the potential assumptions made, other approaches, such as those based on machine learning, were not included. 4) Our model assumes a relatively homogeneous contact pattern within main social settings (households, school, workplace and community) and does not account for potential superspreading events or age-specific susceptibility, which could influence the observed variance of the serial interval. In addition, the designed monitoring scenarios for data collection are also worth to be investigated further, such that we can be actively monitor by testing the sample population, instead of symptom-based household monitoring, etc. These aspects can be particularly relevant for real surveillance.

In conclusion, we carried out a systematic comparison of the widely adopted methodologies for the estimation of SI in a controlled setting for which the ground truth is known. Our results presented that the SI estimating methods employing the contact tracing data or household follow up data outperform the methods only using the aggregated data. The additional sensitive results show that the parameters regulating the epidemic transmission increasing the number of available infector-infectee pairs would improve the effectiveness of SI estimating methods. Our findings highlight the importance of including more available information related to the monitoring data for the design of SI estimating methods.

CRediT authorship contribution statement

Lanlan Yu: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Yunyi Cai:** Writing – review & editing, Methodology, Data curation. **Gui-Quan Sun:** Writing – review & editing, Investigation, Formal analysis. **Jiancheng Lv:** Writing – review & editing, Supervision, Formal analysis. **Quan-Hui Liu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The synthetic population is available from the corresponding author on reasonable request. Other data and codes used in the study are provided in Supplementary Information.

Funding

This work was supported by the National Natural Science Foundation of China (No. 62373264), and the 111 Project under grant agreement B21044.

Conflict of interest

The authors declare that they have no competing interests.

Acknowledgements

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2026.03.005>.

References

- Ajelli, M., Merler, S., Fumanelli, L., Pastore Y Piontti, A., Dean, N. E., Longini, I. M., Halloran, M. E., & Vespignani, A. (2016). Spatiotemporal dynamics of the Ebola epidemic in Guinea and implications for vaccination and disease elimination: A computational modeling analysis. *BMC Medicine*, 14(1), 130. <https://doi.org/10.1186/s12916-016-0678-3>
- Ali, S. T., Wang, L., Lau, E. H. Y., Xu, X.-K., Du, X., Wu, Y., Leung, G. M., & Cowling, B. J. (2020). Serial interval of SARS-CoV-2 was shortened over time by nonpharmaceutical interventions. *Science*, 369(6507), 1106–1109. <https://doi.org/10.1126/science.abc9004>
- Archer, B. N., Tempia, S., White, L. F., Pagano, M., & Cohen, C. (2012). Reproductive number and serial interval of the first wave of influenza A (H1N1) pdm09 virus in South Africa. *PLoS One*, 7(11), Article e49482.
- Cauchemez, S., Donnelly, C. A., Reed, C., Ghani, A. C., Fraser, C., Kent, C. K., Finelli, L., & Ferguson, N. M. (2009). Household transmission of 2009 pandemic influenza A (H1N1) virus in the United States. *New England Journal of Medicine*, 361(27), 2619–2627. <https://doi.org/10.1056/NEJMoa0905498>
- Chen, D., Lau, Y.-C., Xu, X.-K., Wang, L., Du, X., Tsang, T. K., Wu, P., Lau, E. H., Wallinga, J., & Cowling, B. J. (2022). Inferring time-varying generation time, serial interval, and incubation period distributions for COVID-19. *Nature Communications*, 13(1), 7727.
- Cowling, B. J., Fang, V. J., Riley, S., Peiris, J. S. M., & Leung, G. M. (2009). Estimation of the serial interval of influenza. *Epidemiology*, 20(3), 344–347.
- Dai, C., Zhou, D., Gao, B., & Wang, K. (2023). A new method for the joint estimation of instantaneous reproductive number and serial interval during epidemics. *PLoS Computational Biology*, 19(3), Article e1011021.
- Dhiman, A., Yom-Tov, E., Pellis, L., Edelstein, M., Pebody, R., Hayward, A., House, T., Finnie, T., Guzman, D., Lamos, V., & Cox, I. J. (2024). Estimating the household secondary attack rate and serial interval of COVID-19 using social media. *npj Digital Medicine*, 7(1), 1–10. <https://doi.org/10.1038/s41746-024-01160-2>
- Du, Z., Xu, X., Wu, Y., Wang, L., Cowling, B. J., & Meyers, L. A. (2020). Serial interval of COVID-19 among publicly reported confirmed cases. *Emerging Infectious Diseases*, 26(6), 1341.
- Ferguson, N. M., Cummings, D. A., Cauchemez, S., Fraser, C., Riley, S., Meeyai, A., Iamsirithaworn, S., & Burke, D. S. (2005). Strategies for containing an emerging influenza pandemic in Southeast Asia. *Nature*, 437(7056), 209–214.
- Forsberg White, L., & Pagano, M. (2008). A likelihood-based method for real-time estimation of the serial interval and reproductive number of an epidemic. *Statistics in Medicine*, 27(16), 2999–3016. <https://doi.org/10.1002/sim.3136>
- Fraser, C., Donnelly, C. A., Cauchemez, S., Hanage, W. P., Van Kerkhove, M. D., Hollingsworth, T. D., Griffin, J., Baggaley, R. F., Jenkins, H. E., Lyons, E. J., Jombart, T., Hinsley, W. R., Grassly, N. C., Balloux, F., Ghani, A. C., Ferguson, N. M., Rambaut, A., Pybus, O. G., & Lopez-Gatell, H. (2009). Pandemic potential of a strain of influenza A (H1N1): Early findings. *Science*, 324(5934), 1557–1561. <https://doi.org/10.1126/science.1176062>
- He, X., Lau, E. H., Wu, P., Deng, X., Wang, J., Hao, X., Lau, Y. C., Wong, J. Y., Guan, Y., & Tan, X. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nature Medicine*, 26(5), 672–675.
- Kenah, E., Lipsitch, M., & Robins, J. M. (2008). Generation interval contraction and epidemic data analysis. *Mathematical Biosciences*, 213(1), 71–79.
- Lee, H., Lee, S., & Lee, C. H. (2016). Stochastic methods for epidemic models: An application to the 2009 H1N1 influenza outbreak in Korea. *Applied Mathematics and Computation*, 286, 232–249.
- Lessler, J., Reich, N. G., & Cummings, D. A. T. (2009). Outbreak of 2009 Pandemic Influenza A (H1N1) at a New York City School. *New England Journal of Medicine*, 361(27), 2628–2636. <https://doi.org/10.1056/NEJMoa0906089>
- Leung, Y. H., Li, M. P., & Chuang, S. K. (2011). A school outbreak of pandemic (H1N1) 2009 infection: Assessment of secondary household transmission and the protective role of oseltamivir. *Epidemiology and Infection*, 139(1), 41–44.
- Levy, J. W., Cowling, B. J., Simmerman, J. M., Olsen, S. J., Fang, V. J., Santarattiwong, P., Jarman, R. G., Klick, B., & Chotipitayasunondh, T. (2013). The serial intervals of seasonal and pandemic influenza viruses in households in Bangkok, Thailand. *American Journal of Epidemiology*, 177(12), 1443–1451.
- Li, M., Liu, K., Song, Y., Wang, M., & Wu, J. (2021). Serial interval and generation interval for imported and local infectors, respectively, estimated using reported contact-tracing data of COVID-19 in China. *Frontiers in Public Health*, 8, Article 577431.
- Liu, Q.-H., Ajelli, M., Aleta, A., Merler, S., Moreno, Y., & Vespignani, A. (2018). Measurability of the epidemic reproduction number in data-driven contact networks. *Proceedings of the National Academy of Sciences*, 115(50), 12680–12685. <https://doi.org/10.1073/pnas.1811115115>
- Liu, Q.-H., Zhang, J., Peng, C., Litvinova, M., Huang, S., Poletti, P., Trentini, F., Guzzetta, G., Marziano, V., & Zhou, T. (2022). Model-based evaluation of alternative reactive class closure strategies against COVID-19. *Nature Communications*, 13(1), 322.
- Madewell, Z. J., Yang, Y., Longini, I. M., Halloran, M. E., Vespignani, A., & Dean, N. E. (2023). Rapid review and meta-analysis of serial intervals for SARS-CoV-2 Delta and Omicron variants. *BMC Infectious Diseases*, 23(1), 429. <https://doi.org/10.1186/s12879-023-08407-5>
- Merler, S., & Ajelli, M. (2010). The role of population heterogeneity and human mobility in the spread of pandemic influenza. *Proceedings of the Royal Society B: Biological Sciences*, 277(1681), 557–565. <https://doi.org/10.1098/rspb.2009.1605>
- Merler, S., Ajelli, M., Camilloni, B., Puzelli, S., Bella, A., Rota, M. C., Tozzi, A. E., Muraca, M., Meledandri, M., & Iorio, A. M. (2013). Pandemic influenza A/H1N1pdm in Italy: Age, risk and population susceptibility. *PLoS One*, 8(10), Article e74785.
- Nash, R. K., Bhatt, S., Cori, A., & Nouvellet, P. (2023). Estimating the epidemic reproduction number from temporally aggregated incidence data: A statistical modelling approach and software tool. *PLoS Computational Biology*, 19(8), Article e1011439.
- Nishiura, H., Linton, N. M., & Akhmetzhanov, A. R. (2020). Serial interval of novel coronavirus (COVID-19) infections. *International Journal of Infectious Diseases*, 93, 284–286.
- Orellano, P. W., Reynoso, J. I., Grassi, A., Palmieri, A., Uez, O., & Carlino, O. (2012). Estimation of the serial interval for pandemic influenza (pH1N1) in the most southern province of Argentina. *Iranian Journal of Public Health*, 41(12), 26.
- Paquet, C., Coulombier, D., Kaiser, R., & Ciotti, M. (2006). Epidemic intelligence: A new framework for strengthening disease surveillance in Europe. *Euro Surveillance*, 11(12), 5–6. <https://doi.org/10.2807/esm.11.12.00665-en>
- Park, S. W., Akhmetzhanov, A. R., Charniga, K., Cori, A., Davies, N. G., Dushoff, J., Funk, S., Gostic, K., Grenfell, B., Linton, N. M., Lipsitch, M., Lison, A., Overton, C. E., Ward, T., & Abbott, S. (2024). Estimating epidemiological delay distributions for infectious diseases. *medRxiv*, 2024. <https://doi.org/10.1101/2024.01.12.24301247>, 01.12.24301247.

- Park, M., Cook, A. R., Lim, J. T., Sun, Y., & Dickens, B. L. (2020). A systematic review of COVID-19 epidemiology based on current evidence. *Journal of Clinical Medicine*, 9(4), 967.
- Park, S. W., Sun, K., Champredon, D., Li, M., Bolker, B. M., Earn, D. J. D., Weitz, J. S., Grenfell, B. T., & Dushoff, J. (2021). Forward-looking serial intervals correctly link epidemic growth to reproduction numbers. *Proceedings of the National Academy of Sciences*, 118(2), Article e2011548118. <https://doi.org/10.1073/pnas.2011548118>
- Rai, B., Shukla, A., & Dwivedi, L. K. (2021). Estimates of serial interval for COVID-19: A systematic review and meta-analysis. *Clinical Epidemiology and Global Health*, 9, 157–161.
- Savage, R., Whelan, M., Johnson, I., Rea, E., LaFreniere, M., Rosella, L. C., Lam, F., Badiani, T., Winter, A.-L., Carr, D. J., Frenette, C., Horn, M., Dooling, K., Varia, M., Holt, A.-M., Sunil, V., Grift, C., Paget, E., & King, M. (2011). Assessing secondary attack rates among household contacts at the beginning of the influenza A (H1N1) pandemic in Ontario, Canada, April–June 2009: A prospective, observational study. *BMC Public Health*, 11(1), 234. <https://doi.org/10.1186/1471-2458-11-234>
- Susvitasari, K., Tupper, P., Stockdale, J. E., & Colijn, C. (2023). A method to estimate the serial interval distribution under partially-sampled data. *Epidemics*, 45, Article 100733.
- te Beest, D. E., Wallinga, J., Donker, T., & van Boven, M. (2013). Estimating the generation interval of influenza A (H1N1) in a range of social settings. *Epidemiology*, 24(2), 244–250.
- Tuite, A. R., Greer, A. L., Whelan, M., Winter, A.-L., Lee, B., Yan, P., Wu, J., Moghadas, S., Buckeridge, D., & Pourbohloul, B. (2010). Estimated epidemiologic parameters and morbidity associated with pandemic H1N1 influenza. *Canadian Medical Association Journal*, 182(2), 131–136.
- Vink, M. A., Bootsma, M. C. J., & Wallinga, J. (2014). Serial intervals of respiratory infectious diseases: A systematic review and analysis. *American Journal of Epidemiology*, 180(9), 865–875.
- White, L. F., Wallinga, J., Finelli, L., Reed, C., Riley, S., Lipsitch, M., & Pagano, M. (2009). Estimation of the reproductive number and the serial interval in early phase of the 2009 influenza A/H1N1 pandemic in the USA. *Influenza and Other Respiratory Viruses*, 3(6), 267–276. <https://doi.org/10.1111/j.1750-2659.2009.00106.x>
- WHO Ebola Response Team, Aylward, B., Barboza, P., Bawo, L., Bertherat, E., Bilivogui, P., Blake, I., Brennan, R., Briand, S., Chakauya, J. M., Chitala, K., Conteh, R. M., Cori, A., Croisier, A., Dangou, J. M., Diallo, B., Donnelly, C. A., Dye, C., & Eckmanns, T. (2014). Ebola virus disease in West Africa—The first 9 months of the epidemic and forward projections. *New England Journal of Medicine*, 371(16), 1481–1495. <https://doi.org/10.1056/NEJMoa1411100>
- Xiong, S., Cai, S., Wei, F., Chen, G., Zheng, K., & Xie, J. (2024). Estimation methods of reproduction numbers for epidemics of varying strains of COVID-19. *Journal of Biosafety and Biosecurity*, 6(4), 265–270. <https://doi.org/10.1016/j.jobbb.2024.10.003>
- Yang, L., Dai, J., Zhao, J., Wang, Y., Deng, P., & Wang, J. (2020). Estimation of incubation period and serial interval of COVID-19: Analysis of 178 cases and 131 transmission chains in Hubei province, China. *Epidemiology and Infection*, 148, Article e117.